

Agile Procurement and Dynamic Value for Money to Facilitate Agile Software Projects

Diane Jamieson
Curtin Business School
Curtin University of Technology
Perth, Australia
Diane.Jamieson@cbs.curtin.edu.au

Kevin Vinsen
ADI Limited
Perth, Australia
Kevin.Vinsen@adi-limited.com

Guy Callender
Curtin Business School
Curtin University of Technology
Perth, Australia
Guy.Callender@cbs.curtin.edu.au

Abstract

In situations where software systems are developed under formal contractual arrangements it is common for the buyer and supplier to dispute whether value for money is being delivered, even though it is unlikely that either party will be capable of measuring this outcome.

This problem is exacerbated by a basic incompatibility between the buyer's linear or sequential procurement process and the supplier's agile software development methodology. At the commencement of a software development contract it is probable that a supplier will have accepted terms and conditions relating to scope and price that are philosophically in opposition to agile principles, particularly in respect of an iterative elaboration of requirements. A range of suboptimal project outcomes, including budget and schedule blowouts, typically result in these situations.

This paper proposes an agile procurement process that can operate in conjunction with an agile software development methodology to improve many aspects of software project delivery. This process is supported by a mechanism designed to give the buyer and supplier the ability to define and agree the value for money proposition at various milestones within the project – dynamic value for money.

stem from the relationship between the customer and supplier. Other problems arise when decisions are made using imperfect information and with poor definition of final desired outcomes [3].

Large software development projects are frequently undertaken within a competitive tendering and contracting (procurement) process that creates a formal contractual relationship between separate buyer and supplier organizations, subsequent to the supplier being chosen as being most likely to deliver value for money (VfM). To complicate matters the end user of the software system may even be found in a third organization.

The typical procurement process is linear or sequential in nature (often described as a waterfall approach), with the supplier being engaged after several of the process steps have been completed. The waterfall process forces decisions to be made on scope and budget early in the process, when information is limited. In this situation, the accuracy and appropriateness of the VfM decision is also highly dependent upon the ability of the potential suppliers to formulate a solution in an environment in which probity constraints, particularly for government, minimise the interaction that can occur to confirm the requirements. A range of cost estimation methodologies has evolved to assist in the understanding of requirements and their costs. However software projects regularly exceed budget expectations and it is likely that VfM decisions are made using flawed assumptions, even if the price estimation has been accurate. Certainly there is no generally available means of measuring VfM within a project.

Software development practitioners have embraced agile and iterative techniques in an endeavour to minimise problems associated with the specification of requirements and scope creep [4]. However the current waterfall procurement process prevents suppliers from being formally engaged at early decision points. As a

1. Introduction

A review of studies of software project problems by Moløkken and Jørgensen's [1] supports the view expressed in the Chaos Report [2] that a number of specific problems are routinely identified. These include: poor requirements specification, inadequate user involvement, changing requirements, lack of executive sponsorship, poor project management and inaccurate cost estimates. Some of these problems (such as executive sponsorship and user involvement)

result the power of an agile development methodology cannot be fully realised.

This paper proposes an agile procurement process that formalises greater interaction between a buyer and potential suppliers early in the procurement process, and is compatible with an agile software development approach. This agile procurement process can also provide the basis upon which to build a Dynamic VfM model for determining and measuring VfM outcomes at various stages in a project.

2. Issues with Waterfall Procurement

2.1. A Range of Disciplines

The procurement of software development involves a number of different disciplines (such as accounting, finance, law, management and Information Technology) and associated methodologies in the early phases, before the winning supplier has been engaged. This leads to a range of ongoing difficulties. Those areas most likely to be associated with improved software project outcomes within an agile procurement process are discussed below.

2.2. The Procurement Process

Figure 1 shows a general representation of the waterfall procurement process [5], which has been previously described in detail [6].

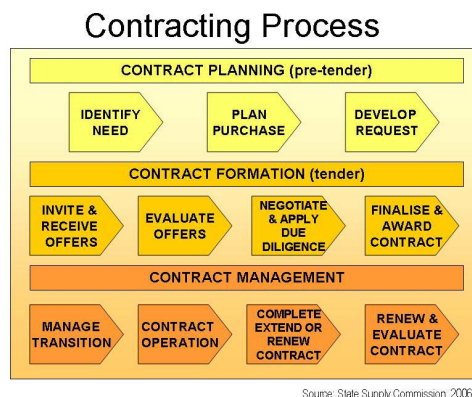


Figure 1

In the early phases of a software development project strict probity rules, which dictate how a buyer and potential suppliers should communicate during the tendering process, often restrict the opportunity for an extensive dialogue which otherwise would uncover the full extent of the project. The buyer's VfM decision is based upon the tenderers interpretation of the requirements and the winning tenderer is unlikely to

have an appreciable understanding of the task ahead by the contract award phase. Thus the resulting contract will contain terms and conditions that inappropriately reflect the outcomes that will eventually required to be delivered.

2.3. Software Cost Estimation and the Winner's Curse

One effect of the significant proportion of software projects that have cost overruns [1] is the frequently expressed view that software cost estimation is over optimistic [7]. However anecdotal evidence would suggest that even when the predicted cost proves to be fairly accurate, competing budget priorities frequently encourage customers to allocate insufficient budget.

The procurement process inherently contributes to this problem through encouraging suppliers to be optimistic in their price estimates in order to win the tendering competition [8]. Even though most procurers profess to use some form of VfM evaluation, rather than lowest price, it is rare for a supplier in a higher price band to win. Buyers instinctively use competition as a price reduction strategy (as exemplified by early experiences of e-auctions). The resulting supplier optimism or "winner's curse" [8] has frequently been known to cause low or negative supplier profits [9], [8] and can also harm the buyer [8].

2.4. Requirements Specification

Consultants are often used in the early phases to help produce a requirements specification to be supplied with the tender documentation. However a consultant's view, which is a form of expert opinion and is provided before the development process has begun, may not address all the potential complexities that will be uncovered by the vendor organization which is eventually required to deliver the system. The iterative and agile model often applied to contemporary software development recognises that requirements will continue to evolve and be discovered during the development phase and that continual reprioritising of requirements will lead to greater customer satisfaction for a lower price [4].

Project management practice is also moving towards more agile approaches [10], with one driver being more effective requirements elicitation, and to support this shift in practice a similar change in the procurement process would be helpful.

2.5. The Value for Money Decision

The Commonwealth of Australia [11] describes VfM as the core principle underpinning procurement and “requires a comparative analysis of *all* relevant costs and benefits of each proposal through the whole procurement cycle (whole of life costing)”. It should include consideration for “the maturity of the market for the property or service sought; the performance history of each prospective supplier; the relative risk of each proposal; the flexibility to adapt to possible change over the lifecycle of the property or service; financial considerations including all relevant direct and indirect benefits and costs over the whole procurement cycle; the anticipated price that could be obtained, or cost that may be incurred, at the point of disposal; and the evaluation of contract options (for example, contract extension options)”.

This description, which is very subjective, includes elements relating to the type of procurement process to be undertaken, the type of good or service to be sourced, and the type of supplier to be selected. It is a multi-phase activity, and the choice of the winning tenderer is the principal VfM decision acknowledged in the process. There is often no reference to the continuing evaluation of VfM during the contract, although Key Performance Indicators (KPIs) represent a common, though often imperfect, contract management tool and imply that some element of VfM assessment is being undertaken.

For software development procurement the VfM decision that selects the winning tenderer traditionally includes an estimation of the potential management and technical capability of the supplier, the technical and implementation solutions offered, and the price. The VfM decision usually results from an examination of the ranking of the suppliers in relation to these qualitative criteria and a separate ranking of their normalised prices. The subjective nature of the process means that the result may not be reproducible with different evaluators, an outcome that creates significant uncertainty about the reliability of the decision process.

2.6. Value for Money Measurement

The subjective nature of a typical VfM decision creates a common area of contention between buyers and suppliers during the contract management phases of the procurement. Buyers are likely to believe that suppliers are not delivering VfM and suppliers are potentially frustrated by the buyer's freedom to contest the VfM received from the supplier even though there are no generally agreed means to monitor VfM.

Furthermore, although VfM is frequently described [11] it is not defined in terms of generally applicable principles [6]. This lack of general principles means that during a software development contract there will be an absence of a regular evaluation of VfM. Project management techniques, such as earned value management, can be used to monitor performance and KPIs are typically used in the implementation phases of systems, although they are difficult to define and measure at any stage of the development phases.

3. An Agile Procurement Process is Necessary

A research project being undertaken in Australia by the authors is aimed at *optimising VfM in software development through standardised cost metrics in the procurement, development and project management phases*.

It is becoming increasingly clear that a significant revamp of the procurement process is one way to contribute to this objective. However, there are still significant problems with software cost estimation, as highlighted by Gruschke and Jørgensen [12], who believe that software cost estimation results have not improved in twenty years despite extensive developments in formal estimation methodologies. It seems as though methodologies cannot adapt as quickly as new issues arise. Thus a new procurement process that reduces the impact of inaccuracies in software cost estimation is one way of approaching this challenge. This will have the added benefit of facilitating the requirements engineering and cost estimation processes.

Secondly a technique for measuring VfM at many stages in a project, not only at the commencement, is highly desirable. However this approach is difficult to incorporate in the current procurement process, particularly because project budgets are constrained by traditions relating to the setting of “whole of project” cost estimates and the matching of costs to financial reporting principles. Nevertheless, the merits of an agile procurement process, designed to match the agile methodologies increasingly associated with software development and project management are worth serious consideration.

4. Agile Procurement to Support Dynamic VfM

4.1. A Proposed Agile Model

The search for a mechanism for monitoring Dynamic VfM has resulted in a preliminary approach

that anticipates the adoption of an agile procurement model, and incorporates a modified approach to software cost estimation and facilitation of requirements engineering activities. Figure 2 below depicts the modified procurement process in which a dynamic VFM model could operate.

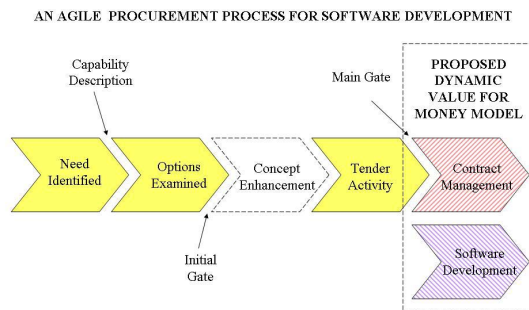


Figure 2

The solid boxes in Figure 2 show a simplified representation of the typical waterfall procurement process for software development being generally undertaken by a significant proportion of organizations (and adapted from Figure 1). The Concept Enhancement box represents an additional step being proposed by the authors to improve the understanding of the requirements by both the customer and potential suppliers, and to also determine the milestone content for each Dynamic VfM determination. Versions of this step already operate in some jurisdictions, many informally. However this paper documents the incorporation of some of these, and potentially new, principles. The proposed Dynamic VfM model discussed in Section 5 would operate in the existing contract management and system development phases.

Capability Description in Figure 2 represents the formal identification of the proposed capability and its broad expected cost. *Initial Gate* is the term used by the UK Government [13] to represent approval to procure and *Main Gate* signifies approval to contract with the winning tenderer.

4.2. Current Concept Enhancement Approaches

Currently procurers have a number of techniques operating as forms of “Concept Enhancement”, three of which are discussed below.

Expression of Interest (EOI) [14] is a method of obtaining feedback from industry on proposed activities.

One of the authors was employed in the 1990s on a large Project Definition Study (PDS) for the UK Ministry of Defence in which two suppliers were paid to develop complex technological solutions to a certain point before a Best and Final Offer (BAFO) process determined the supplier chosen to deliver the final system. The best concepts from both solutions developed in the PDS were available for inclusion in the final design.

A recent Australian Government Request for Tender (RFT) document [15] proposed three possible outcomes from the tender process:

- The tender process is terminated,
- A contract is entered into with the preferred tenderer to develop the detailed design specification, which will be used in a subsequent tender process, or
- Contracts are entered into with two or more competing suppliers to develop the detailed design specification and then one of these suppliers is chosen to develop the final system without any further tender process.

The second possible outcome above represents the use of potential suppliers, rather than consultants, to develop the specification and documentation for the Tender Activity Phase shown in Figure 2. The third possible outcome above is equivalent to a PDS. Both of these methods involve expenditure of part of the project budget as a “Concept Enhancement” type activity, which is called a “Design Contest” in the above mentioned RFT.

4.3. Derisking Using Concept Enhancement

In 2005 the UK Ministry of Defence [13, p. 33] “set a target figure for 15% of the total procurement costs to be invested pre-Main Gate and the bulk of this would be spent on derisking”. In Australia the Defence Procurement Review 2003 [16] recommended a similar approach, which it considered to be useful for most organizations sourcing complex services. Concept Enhancement in Figure 2 represents a key activity that could occur after the preferred procurement option has been defined and agreed at Initial Gate and consume a proportion of the 15% of total procurement costs with activities that involve potential suppliers, rather than consultants.

The Unified Process (UP) [4] offers a potential model for adaptation for this purpose. UP suggests running projects in two contract phases, with the first phase being a relatively short fixed-term and fixed-price contract. The goal of this first step is to complete the first few iterations of the project, undertake evolutionary requirements analysis and produce

preliminary software to be provided to tenderers for a phase two fixed-price contract. The evolutionary refinement of specifications and code in phase one provides higher quality data for phase two estimators. However UP tends to be used by suppliers, not by procurers, and large single phase projects remain the norm.

4.4. A Potential Concept Enhancement Activity

A potential alternative model offered by the authors for the Concept Enhancement phase could be an interim tender activity to identify suppliers to better define the requirements and estimate the cost of the required system, and create a shortlist of suppliers from which the final selection would be made. Tenderers would need to provide an indication of their general capacity to build the final system and would subsequently be advised that the final tender would only be released to suppliers chosen from this Concept Enhancement shortlist (of perhaps three or four suppliers). Suppliers would be paid to develop items such as a proof of concept, preliminary design, preliminary requirements documentation and Dynamic VfM milestones.

The proposed Concept Enhancement phase would combine a number of current procurement options into a new format that should improve the understanding of requirements and their costs and define the interim project milestones for inclusion in the tender documentation. Traditional supplier hesitation in sharing IP should be minimised because the suppliers would be paid for the service, with the most capable supplier having its concepts incorporated in the tender documentation, and less capable suppliers would have the opportunity to learn from the process. This would assist in improving general capability in the supplier community, which should, in turn, benefit the customer community.

4.5. Changes to Cost Estimation Practices with Agile Procurement

Jørgensen [17] claims that expert judgment is the most common cost estimation approach currently in use and that there is no substantial empirical evidence favouring formal estimation techniques. The budgeting approach and structure of contract described above may reduce the criticality of accurate long term cost estimates and instead may create a structured environment for more effectively applying a wide-band Delphi type approach to cost estimation, including expert judgment, using customer and supplier representatives.

5. The Search for a Dynamic VfM Model

A traditional VfM analysis represents the customer's perception of the capability of the supplier to deliver particular outcomes of value to the buyer measured against a number of quantitative and qualitative criteria. The determination of VfM within a project requires a more balanced approach, since at any stage in a project the ability of the supplier to deliver is also influenced by the capability and responsiveness of the customer. The studies discussed in Section 1 have recognised that customer sponsorship (especially at the executive management level) and end-user involvement are crucial to project success. As a consequence it can be argued that any VfM analysis that does not take account of these factors would be incomplete and potentially unfair on the supplier (and also the buyer). This was one of the first principles to be considered in the model, termed "Dynamic VfM" shown in Figure 3 below. This concept has a number of key components – an evolving Dynamic VfM analysis that incorporates both customer and supplier factors, Dynamic VfM decision points (gates) at previously agreed milestones, and an iterative approach to the determination of content and price for each contract phase. This model requires input from an agile procurement process such as that described in Figure 2.

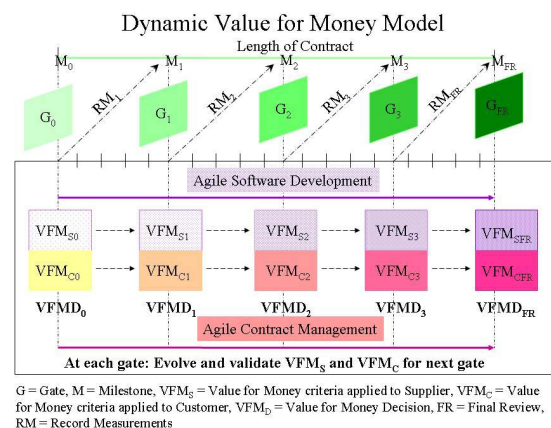


Figure 3

Dynamic VfM is believed to be a term specific to the procurement discipline. However it implies some outcomes that are common to some of the techniques used in other fields, particularly project management. The project management discipline continues to grapple with issues such as success criteria and critical success factors [18], which are also important in procurement projects.

A detailed discussion of the construction of VFM_{Sn} , VFM_{Cn} and $VFMD_n$ as shown in Figure 3 is reserved for another paper.

5.1. Dynamic VfM Components

Although space prevents full elaboration of the VfM model, basic principles can be briefly described. Two VfM factors are crucial in a Dynamic VfM decision within a contract: the performance of the supplier (VFM_S) and the performance of the customer (VFM_C).

VFM_{S0} is the overall score the supplier received as a result of the tender evaluation process. The VFM_{S0} , the evaluation criteria, and the supplier's score should all be available for every contract resulting from a competitive tendering and contracting procurement process. VFM_{C0} is not likely to exist, but could possibly be inferred from information on prior VfM decisions in the procurement process and the type of qualitative selection criteria included by the customer. VFM_{S1} and VFM_{C1} will need to be explicitly defined as one of the first tasks of the initial project iteration. VFM_{Sn} and VFM_{Cn} can then be evolved from VFM_{S1} and VFM_{C1} through incorporation of other relevant factors, such as the critical success factors relevant to each "n". The ability to manage changing requirements and to continue to deliver business value within the project is crucial. "Scoring" or "measuring" in this process is likely to remain subjective. However, provided there is prior agreement between the parties on the factors to be included in the scoring this process has the potential to become a more rigorous contract management tool than those currently applied.

A key advantage of a dynamic VfM process would be the ability to make appropriate changes to improve the performance of a software development contract or the required business outcomes on an agreed basis at agreed times. This would replace the typical adversarial problems that arise over what has been delivered under a traditional software development contract, and how much it has cost (in project management terminology this is the earned value [19], and it is extremely difficult to calculate for incomplete software). However in extreme cases it may also allow the customer to terminate the contract and move to another supplier. Providing the procurement policy process was adapted to include this step, the supplier who originally received the second top VfM score could be approached.

5.2. Determining the Gates

A key issue in a Dynamic VfM model is at what points, or gates, would major VfM determinations be undertaken. One approach would be to use project milestones, although these would need to give the customer a discrete and useful deliverable, not just be milestones that suited the supplier for payment purposes. The Concept Enhancement phase of the agile procurement process described in Section 4 defines a mechanism for determining these gates.

In the manner of UP the specification for the first iteration would be most tightly defined, with more flexibility built into future phases. The customer would own the deliverables and associated intellectual property (IP) from this concept enhancement phase and would feed the most acceptable concepts, including the milestone content of the gates, into the tender documentation for the Tender Activity phase. The Tender Activity phase would result in a fixed price agreement up to the first gate and an agreement to define further fixed prices at each successive gate.

The term "gates" has been adapted from the UK Government's OGC Gateway Process (OGC) [20], which recommends a series of reviews of a procurement activity at five key decisions points. The gates in Figure 3 do not in any way reflect the OGC's gates and will be project specific, rather than the generic procurement phases of the Gateway Process. Nevertheless, the principles have some similarities.

5.3. Content of the Contract

Cost overruns in software development projects partially arise from an inability to accurately estimate the cost of these projects. However, those responsible for collating budget data appear to input these figures into their budget estimates with reasonable confidence, even when evidence shows that this confidence is misplaced. The agile procurement model described above could be expected to reduce the need for accurate long term estimates, as long as it was accepted that the broad estimates defined at the beginning could be progressively enhanced, not only before supplier selection, but also during the course of the contract.

For example, for the contract shown in Figure 2 the length of contract would be the maximum expected time for the development, implementation and maintenance of the system. The budget for this contract would be broadly specified with the most timely information available at the tender phase. However even though the supplier could expect to be retained for the full length of the contract there would

be a guarantee only until Gate 1, at which time a VfM decision would be undertaken on whether to continue. At this stage a fixed price agreement would be determined for Gate 2. This process would be repeatable for further gates.

6. Case Study

A key data source for the Australian study is a regulatory system that comprises an essential link between the community and government. This AUD\$32m system, with 134 use cases and four major releases, was developed over five years for a government department, and contains interfaces to five major systems in other organizations. Although this project did not attempt Dynamic VfM determinations it did undertake a number of the elements described in this paper.

To varying degrees, this project suffered most of the normal project problems, including inadequate user involvement and changing requirements, however the public sector procurement process itself also contributed problems that an agile approach would have addressed.

The agency undertook a sophisticated and sensible approach to the procurement activity within the constraints imposed and had a mature relationship with the supplier. There was a steering committee with representation from a range of customer and supplier business areas, which on a monthly basis recorded risks, problems and potential solutions. These monthly steering committee minutes and the monthly project status reports, along with the results of the original evaluation process, contain a wealth of information for building and testing VFM_{Sn} , VFM_{Cn} and VFM_{Dn} for this project.

The four major releases acted as opportunities to review deliverables and costs and provided break points with tangible and useable deliverables that could still be implemented even if budget or organizational reasons forced delays or even the abandonment of the project. These releases were defined during the period between the selection of the preferred supplier and the formal signing of the contract. During this period the supplier and customer exchanged further information on the requirements, conducted some work to prove the viability of the solution proposed by the supplier and refined the contract to support the staged delivery approach. Suppliers frequently contribute to project outcomes in this manner outside of a contractual relationship in an attempt to reduce delays in project schedule. However this will always leave a supplier at risk until a contract is executed.

The releases were structured with both fixed price and schedule of rates components. In general the application development components were included in the fixed price component and system integration and work on interfaces were completed on a schedule of rates basis. This dual approach was chosen because the mitigation of the application development risk was considered to be under the control of the supplier whereas much of the risk for the system integration and interface work was considered to be under the control of the customer. It was the customer which was also responsible for the negotiation of the interfaces with its partner organisation and the specification of the hardware and software environment for the application.

The work to be included in both the fixed price and scheduled components of each release was defined in a work breakdown structure. The work breakdown structure was formally signed off for the first release prior to the commencement of the contract and a process was put into place to formally review the work breakdown structure prior to the commencement of each subsequent release. The intention was that where changes to the scope of the fixed price components were identified these were to be costed as variations using the schedule of rates and added into the fixed price component.

The project, which provides a rich source of data for contract costing and operational analysis, is now complete and the system is fully functioning.

7. Conclusion

The proposed agile procurement process and Dynamic VfM model have more complexities than can be adequately described in the space available. Additionally, the concepts have not been formally tested. However it is anticipated that adoption of the agile procurement principles outlined in this paper should lead to projects that meet stakeholder expectations more comprehensively.

Discussions with a range of buyer organizations in Australia, including the organization involved in the project comprising the Case Study described above, and suppliers capable of delivering these systems, have identified strong support for the proposed models. One of the largest impediments envisaged in implementing agile procurement across public and private sector organizations is resistance to significant change to existing financial estimations presented by the agile procurement process, and a perception of increased contracting risk, especially amongst those responsible for allocating budgets.

The data is available in the Australian research project to allow agile procurement principles to be tested on a number of large, complete system development procurement projects and permission has been obtained for evaluating projects that are yet to commence. As a consequence there remains much data collection and analysis to be undertaken and hard evidence is some time away. Nevertheless, early trends suggest some amendments to the traditional procurement process will provide significant benefits, especially in relation to large, complex, software development projects.

Indeed, it is safe to say that it will not be possible to resolve all the uncertainties from software development that may arise between any potential buyer and seller. The procurement procedure outlined in this paper is designed to reduce the scale of that uncertainty rather than provide a permanent solution to the problems of software cost and value for money estimations. As the content of the paper reveals, that is a task for another day.

8. References

- [1] K. Moløkken and M. Jørgensen, "A review of surveys on software effort estimation," *Proceedings of International Symposium on Empirical Software Engineering (ISESE'03)*, pp. 223-230, 2003.
- [2] The Standish Group, "The CHAOS Report," http://www.standishgroup.com/sample_research/PDFpages/chaos1994.pdf, 1995.
- [3] Auditor-General of Victoria, "Implementation of RMIT University's Academic Management System," Report on Public Sector Agencies, Melbourne 2003.
- [4] C. Larman, *Agile & iterative development: a manager's guide*. Boston, MA: Pearson Education, Inc, 2004.
- [5] State Supply Commission, "Procurement users' guide," <http://www.ssc.wa.gov.au/procurement04disp.asp?id=1>, 2006.
- [6] D. Jamieson, K. Vinsen, and G. Callender, "Agile procurement: new acquisition approach to agile software development," *Proceedings of 31st Euromicro Conference on Software Engineering and Advanced Applications*, Porto, Portugal, pp. 266-273, 2005.
- [7] D. V. Ferens, "The conundrum of software estimation models," in *Aerospace and Electronic Systems Magazine, IEEE*, vol. 14, 1999, pp. 23-29.
- [8] M. Jørgensen and S. Grimstad, "Over-optimism in software development projects: 'the winner's curse'," *Proceedings of 15th International Conference on Electronics, Communications and Computers (CONIELECOMP 2005)*, pp. 280-285, 2005.
- [9] R. L. Glass, *Software Runaways: Monumental Software Disasters*: Prentice Hall PTR, 1997.
- [10] D. DeCarlo, *eXtreme project management: using leadership, principles, and tools to deliver value in the face of volatility*. San Francisco, CA: Jossey-Bass, 2004.
- [11] Department of Finance and Administration, "Commonwealth procurement guidelines: the principle of value for money," http://www.finance.gov.au/ctc/principle_of_value_for_money.html, 2005.
- [12] T. Gruschke and M. Jørgensen, "Assessing uncertainty of software development effort estimates: the learning from outcome feedback," *Proceedings of 11th IEEE International Metrics Symposium (METRICS 2005)*, pp. 4-13, 2005.
- [13] Ministry of Defence, "The acquisition handbook, edition 6 - October 2005," <http://www.ams.mod.uk/ams/content/handbook/fro nt.htm>, 2005.
- [14] Department of Finance and Administration, "Commonwealth procurement guidelines," <http://www.finance.gov.au/ctc/>, 2005.
- [15] Australian Customs Service, "Request For Tender (RFT) number 05/1094: development of detailed design specification and program management plan for the JOPC Australian Maritime Identification System," 2006.
- [16] M. Kinnaird, "Defence Procurement Review 2003," <http://www.defence.gov.au/publications/dpr180903.pdf>, 2003.
- [17] M. Jørgensen, "Practical guidelines for expert-judgment-based software effort estimation," *IEEE Software*, vol. 22, iss. 3, pp. 57-63, 2005.
- [18] D. Baccarini, "The logical framework method for defining project success," *Project Management Journal*, vol. 30, iss. 4, pp. 25-31, 1999.
- [19] J. R. Meredith and S. J. Mantel, *Project management: a managerial approach*, 5th ed. New York, NY: John Wiley & Sons, 2002.
- [20] Office of Government Commerce, "OGC Gateway Process," www.ogc.gov.uk, 2006.